

Using this result in the remaining two equations from Eq. (7.2.26), we have

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{Bmatrix} \dot{u} \\ \dot{w} \end{Bmatrix} + \begin{bmatrix} 0 & \omega_0 \\ -\omega_0 & 0 \end{bmatrix} \begin{Bmatrix} u \\ w \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

This is a homogeneous system of first-order linear differential equations. At this point students may wish to review the chapter on systems of linear differential equations from their introductory course on ordinary differential equations. Rewriting this system of equations in the usual differential operator notation, we have

$$\begin{bmatrix} D & \omega_0 \\ -\omega_0 & D \end{bmatrix} \begin{Bmatrix} u \\ w \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad (7.2.28)$$

where D is the differential operator. Differentiating the first equation and multiplying the second equation by ω_0 gives

$$\begin{bmatrix} D^2 & \omega_0 D \\ -\omega_0^2 & \omega_0 D \end{bmatrix} \begin{Bmatrix} u \\ w \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

Subtracting the second equation from the first, we have

$$\begin{bmatrix} (D^2 + \omega_0^2) & 0 \\ -\omega_0^2 & \omega_0 D \end{bmatrix} \begin{Bmatrix} u \\ w \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

In a similar manner the system can be further reduced to yield

$$\begin{bmatrix} (D^2 + \omega_0^2) & 0 \\ 0 & (D^2 + \omega_0^2) \end{bmatrix} \begin{Bmatrix} u \\ w \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

We have now reduced a system of two coupled first-order equations to a system of two uncoupled second-order equations. Each of these is easily solved separately to give

$$\begin{Bmatrix} u \\ w \end{Bmatrix} = \begin{Bmatrix} C_1 \sin(\omega_0 t) + C_2 \cos(\omega_0 t) \\ C_3 \sin(\omega_0 t) + C_4 \cos(\omega_0 t) \end{Bmatrix} \quad (7.2.29)$$

Using this result in Eq. (7.2.28), we have

$$\begin{Bmatrix} C_1 \omega_0 \cos(\omega_0 t) - C_2 \omega_0 \sin(\omega_0 t) + C_3 \omega_0 \sin(\omega_0 t) + C_4 \omega_0 \cos(\omega_0 t) \\ -C_1 \omega_0 \sin(\omega_0 t) - C_2 \omega_0 \cos(\omega_0 t) + C_3 \omega_0 \cos(\omega_0 t) - C_4 \omega_0 \sin(\omega_0 t) \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$